

Physiology of the digestive system



Prepared and presented by: Dr. Mazin Salaheldin
Abdalla



WPS Office تعديل من خلال



Swallowing (deglutition)

- Is the process by which solids and/or liquids leave the oral cavity to reach the stomach through the pharynx and esophagus
- **Stages of swallowing :-**
- An (oral) stage: (Voluntary) the initiation of the process
- Pharyngeal stage:(Involuntary), delivers food to the esophagus
- Esophageal stage:(Involuntary). From the esophagus to the stomach



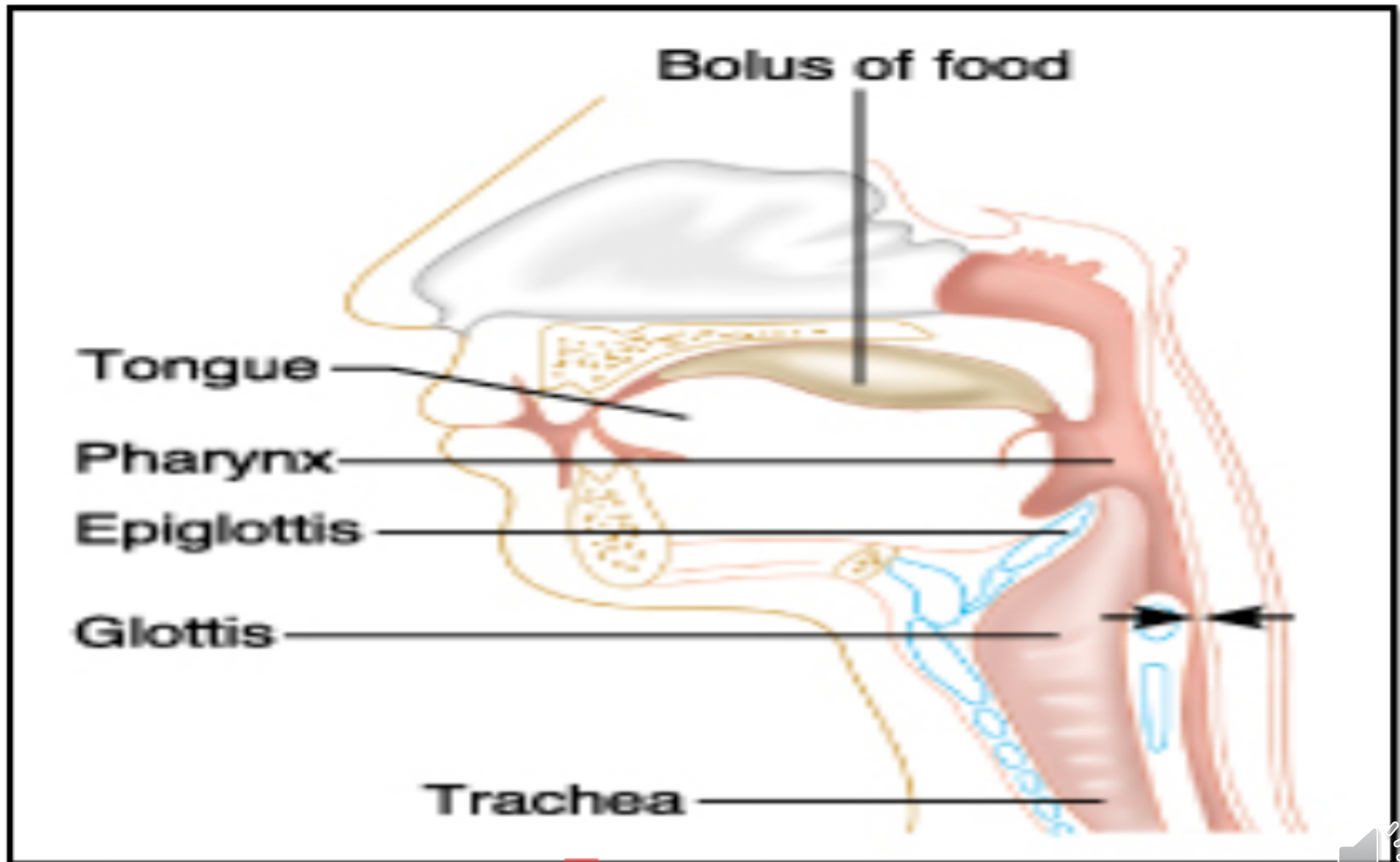
Swallowing (deglutition)

When the food consistency indicates the readiness for swallowing:

- The tongue pressures the food backwards and up, pushing it into the pharynx
- Respiration must be stopped
- The lips and jaw are now closed
- The stage is **voluntary**, however, from then on swallowing cannot be stopped (it becomes automatic)



Phase ONE



The pharyngeal stage

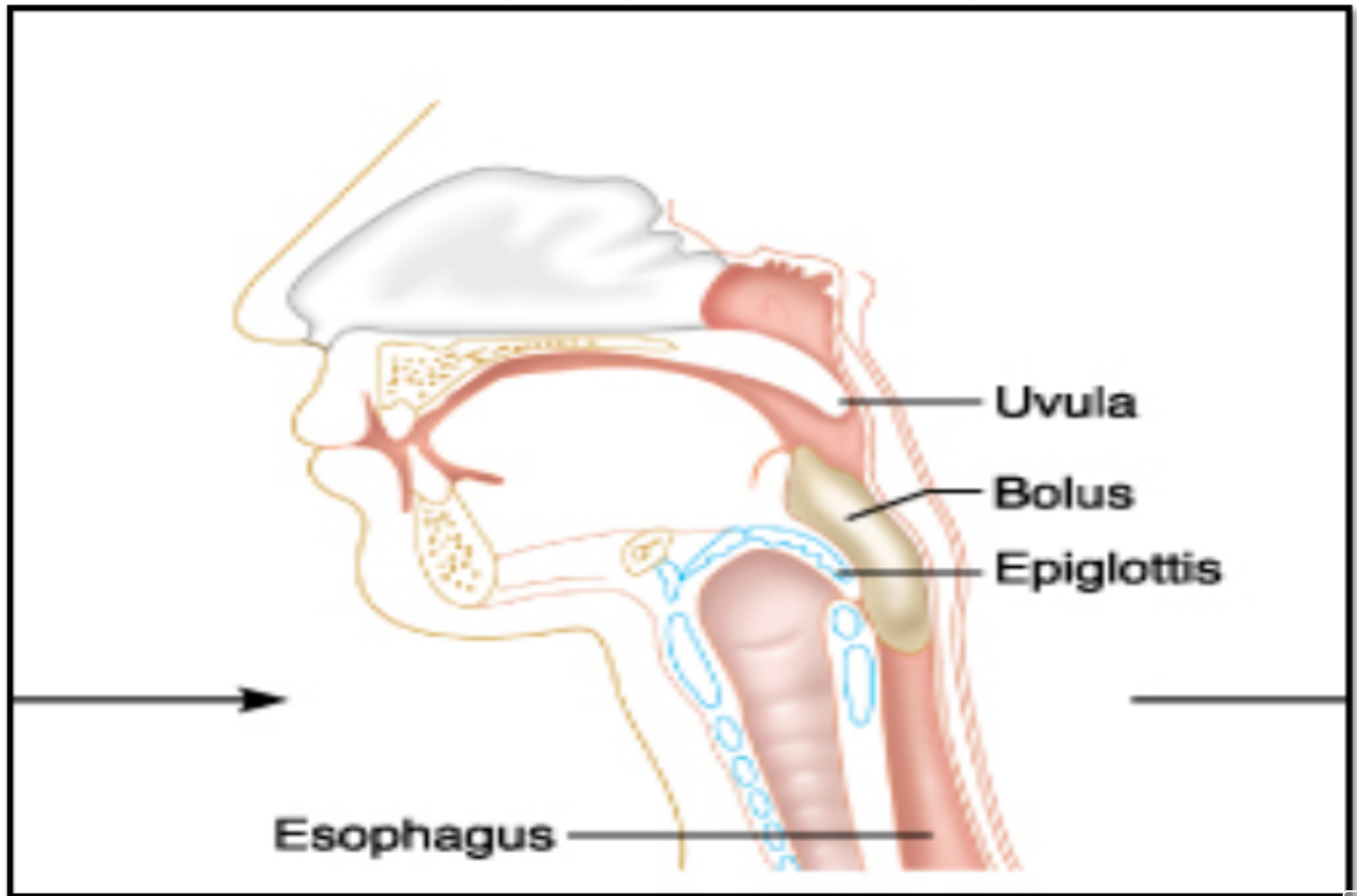
- The bolus is pushed further backwards by the tongue against the epiglottis
- The epiglottis bends over the larynx to cover its opening
- the larynx is closed by contraction of the glottis and approximation of the vocal folds
- the bolus is propelled downwards by contraction of the middle and inferior constrictors the of the pharynx and reaches the upper end of the esophagus



- As soon as the bolus enters the esophagus, the cricopharyngeal sphincter closes, the vocal folds open and the epiglottis returns to its previous position



Pharyngeal stage



What if...

Protective reflexes which prevent entry of food into the air passage include :-

Approximation of palatopharyngeal folds prevent passage of large objects of unmasticated food

Approximation of vocal cords & elevation of the larynx, epiglottis covers & shuts off the glottis



What if....

Cessation (stoppage) of breathing for 2-3 seconds

Opening the way of the bolus into the esophagus by relaxation of upper esophageal sphincter



Nervous control of the pharyngeal stage :-

- Is a reflex act
- It initiated by voluntary movement of food into the back of the mouth
- This stimulates pharyngeal sensory receptors
- Impulses are transmitted through the sensory portion of the trigeminal & glossopharyngeal nerves
- To the swallowing center which is collection of neurons distributed in the medulla & lower Pons



Nervous control of the pharyngeal stage :-

- Motor impulses from swallowing center are transmitted to the pharynx & upper esophagus via the 5th, 9th, 10th, & 12th cranial nerves
- Response coordinated contractions to propel the bolus toward the esophagus, avoiding the larynx, with inhibition of respiration



Esophagus

- Is about 25 cm long, with an upper & lower sphincters
- The wall of the upper third of the esophagus is a skeletal muscle, the middle third is a mixture while the lower third is smooth muscle
- ***Upper esophageal sphincter :-***
- The upper 3 cm of the esophagus
- It is an anatomical sphincter formed by cricopharyngeal muscle & known as the cricopharyngeal sphincter



Esophagus

- Its closed at rest at high pressure (40–100mmHg)
- It open for only one second during swallowing because it prevents entry of air into the esophagus which causes distention of the stomach
- ***lower esophageal sphincter :-***
- The lowest 3 cm of the esophagus known as gastro- esophageal sphincter
- It is a physiological sphincter
- Its an area of high pressure(15–35mmHg)
- Its closed at rest and open for few seconds with swallowing



Esophagus

- Relaxation of the sphincter caused by :
 - Distention of the pharynx by food
 - Vagal stimulation (by releasing VIP)
 - Secretin & CCK
- The sphincter is closed by : -
 - Gastrin hormone
 - Vagus nerve releasing Ach
 - Sympathetic stimulation

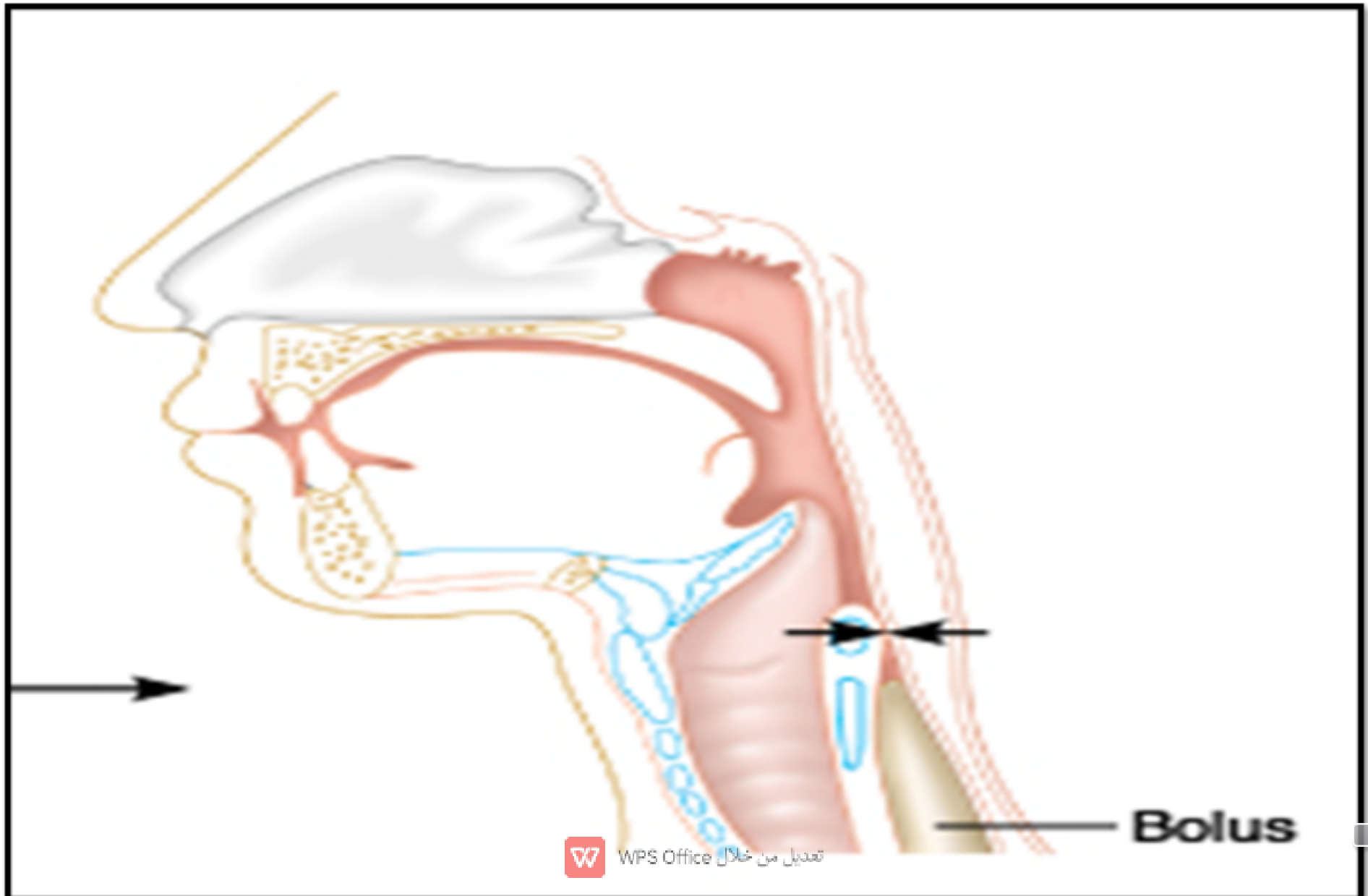


Esophageal stage

- Function of the esophagus is to conduct food rapidly from the pharynx to the stomach and its movements are organized specifically for this function
- There are 2 types of peristaltic movements in the esophagus : ***primary & secondary peristalsis***
- ***In peristalsis*** , stretch of the wall of the esophagus results in contraction behind the bolus (by release of Ach & substance P) and relaxation in front of it (by NO, VIP or ATP)



Esophageal stage



Esophageal stage

- This pushes the bolus downwards and then the process repeats itself
- Primary peristalsis is a continuation of peristaltic wave that begins in the pharynx
- This wave passes all the way from the pharynx to the stomach in about 8- 10 seconds
- If primary peristaltic wave fails to move the food secondary peristaltic wave result from the distension of the esophagus through vago – vagal reflex

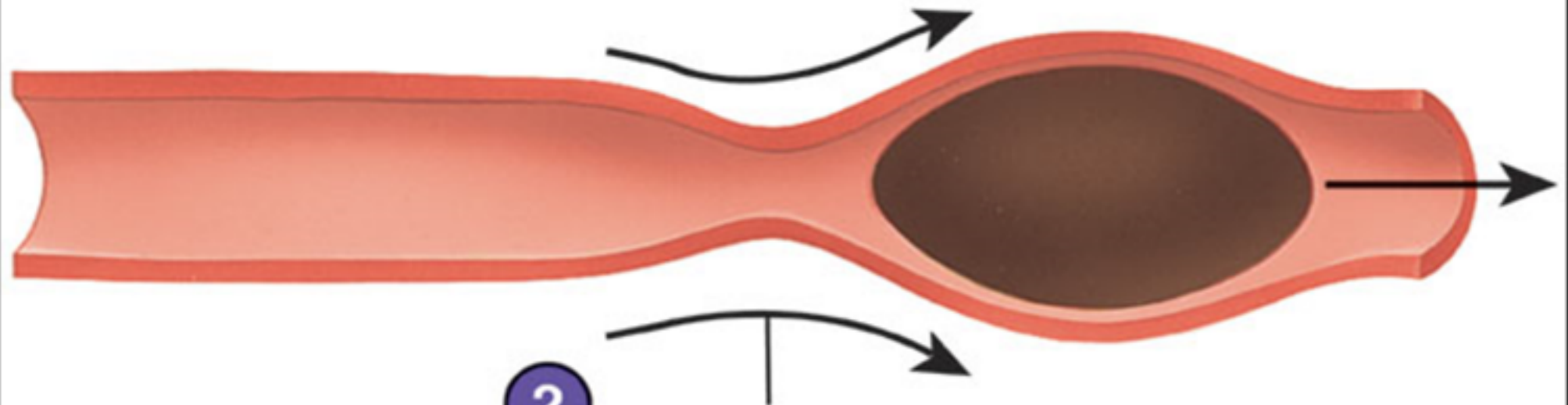
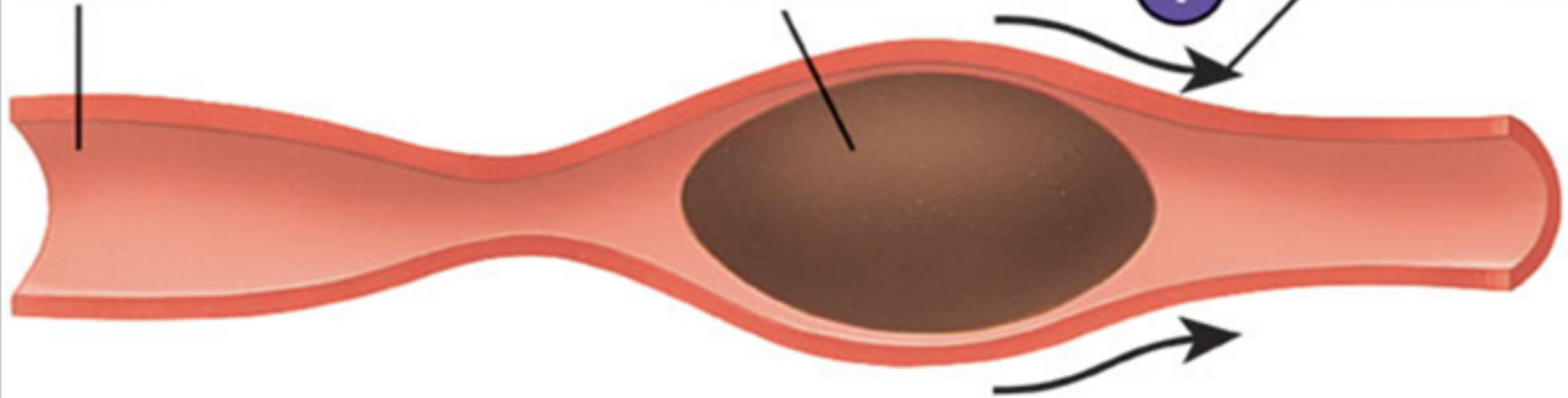


Digestive tract

Bolus or chyme

Wave of relaxation

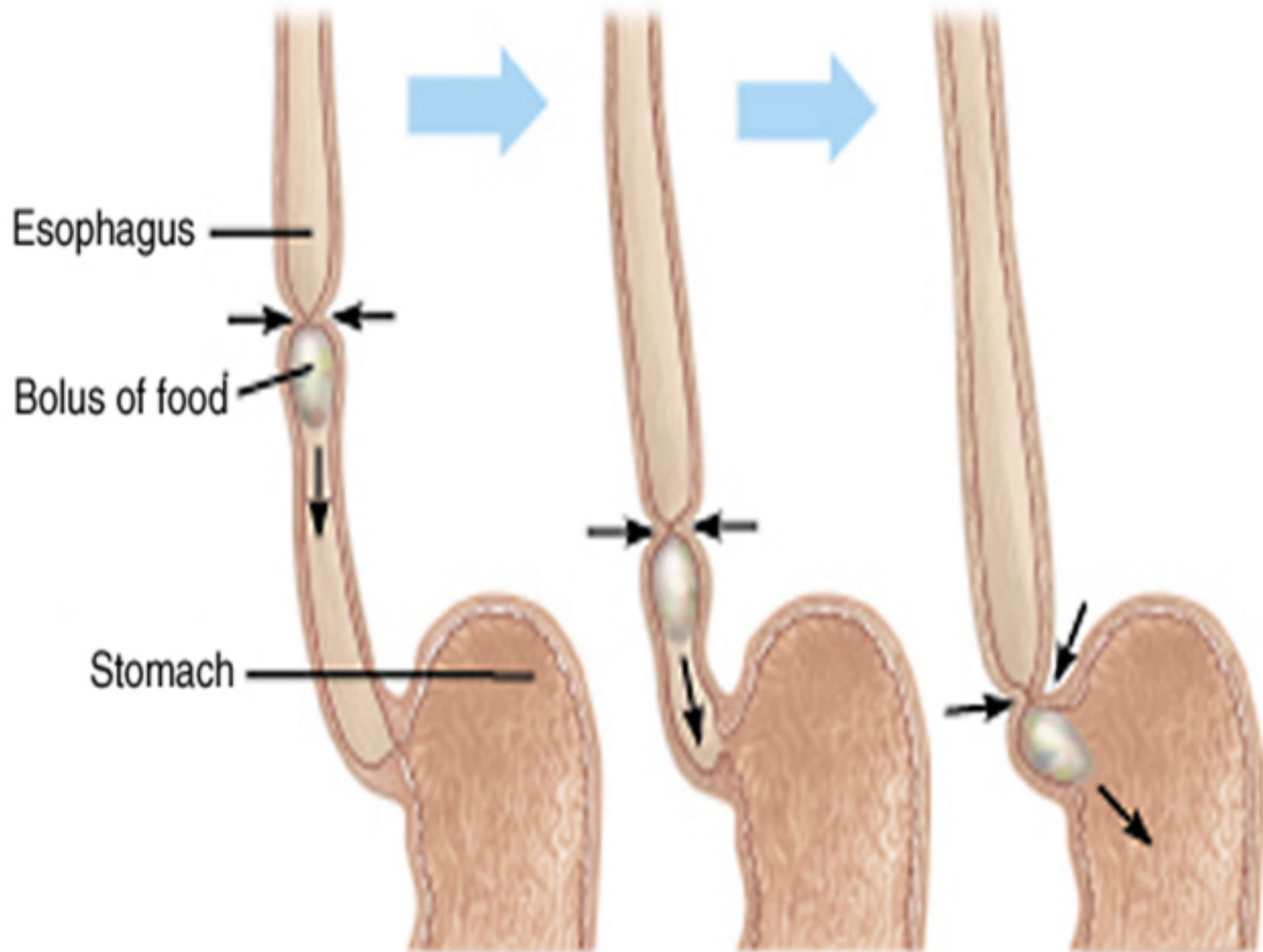
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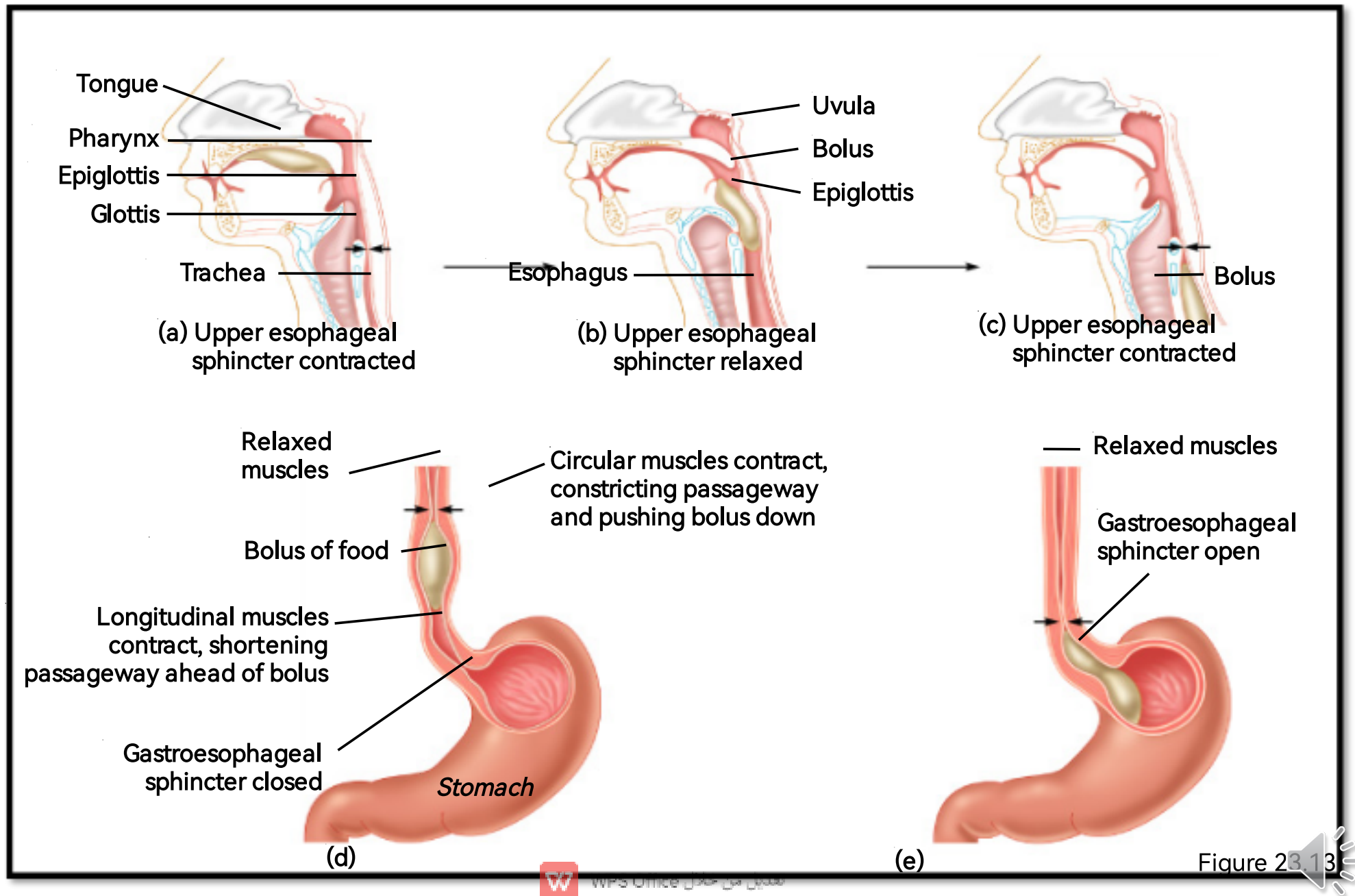
Wave of contraction





(a) Peristalsis

Deglutition. The whole movie



Competence of the lower esophageal sphincter

- LOS is competent in the preventing reflux of gastric contents into the esophagus
- This is due to:
 - 1- Its high resting pressure (15–35)
 - 2- The intra-abdominal location which is compressed from outside by high pressure
 - 3- The crural fibers of the diaphragm, which act as an external sphincter & causing narrowing of the esophagus



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4- The mucosal fold of the stomach, which act as valves that close the esophagus

5- The oblique entry of the esophagus into the stomach, results narrowing of its opening



Next:

Gastric physiology

